



SC: 504 - Computational Lab - I

Test - 3

Sem - I, M.Sc.(Scientific Computing)

Roll-No & Name : MS2513 KAMOD KUNAL

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Time: 10:15 AM to 11:15 AM

Date: September 16, 2025

Max mark: 20

1. Attempt ALL

- (a) Write a C program that takes a negative integer n as input. (4)

Using a while loop, iterate from 1 down to n (inclusive). For each number, display an output based on the following rules:

- If the number is divisible by both 3 and 5, print: "Epic Combo!"
- If the number is divisible by both 2 and 3, print: "FizzTwos!"
- If the number is divisible by 5 "Buzz Deep!"
- If the number is divisible by 3 only, print: "Fizz"
- If the number is divisible by neither 2, 3, nor 5, print: "Chill"

Example : Input: -7

Output:

-1: Chill
-2: Chill
-3: Fizz
-4: Chill
-5: Buzz Deep!
-6: FizzTwos!
-7: Chill

- (b) Write a C program that prints the following pattern using a while loop: (4)

Accept n from the user. For example $n = 5$, the output should be:

1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

- (c) Write a C program that: (6)

1. Reads n integers into a dynamically allocated array.
2. Uses a loop to calculate the average of the elements in the array.
3. Prints the average.

Note : If your roll number is even solve using for loop, else make use of while loop.

Example Input:

5
10 20 30 40 50

Example Output:

The average of the array elements is: 30

(d) Write a C program that:

(6)

1. Reads n integers into a dynamically allocated array.
2. Uses a **for** loop with **if--else if--else** conditions to count how many numbers are:
 - Positive
 - Negative
 - Zero
3. Uses **nested loops** to print all pairs of elements in the array where the sum of the pair is positive.

Example Input:

```
5
3 -1 0 4 -2
```

Example Output:

```
Positive count: 2
Negative count: 2
Zero count: 1
Pairs with positive sum:
(3, -1)
(3, 0)
(3, 4)
(3, -2)
(-1, 4)
(0, 4)
(4, -2)
```

— Why don't programmers like nature? — Too many bugs! —